

Using Predictive Modeling to Manage Employee Attrition

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Introduction: Managing Employee Attrition

Employee retention and the costs associated with replacing workers due to attrition are challenges commonly faced by most employers. In 2025, the annual voluntary turnover rate for all employees in the United States was 23.9% (U.S. Bureau of Labor Statistics 2026, Table E). According to a survey conducted by Gallup, employers report the replacement cost for an employee to be anywhere from 40% to 200% of the employee's annual salary. Furthermore, a survey of employees who recently left their employer indicates that 42% of employee turnover can be prevented (Wigert and Maese 2026). There are many resources that provide advice to help organizations and managers address the reasons employees leave, including offering competitive pay, performance review best practices, and ensuring there are opportunities for advancement. While the publicly available advice from prominent business news and information sources is quite credible, it's unlikely that all of the reasons for turnover are equally applicable to all employees and all businesses. Additionally, an employer may be facing resource constraints and may wish to maximize the impact of any resource expenditure. So how does an employer identify and address the reasons for employee attrition? Can resources be deployed strategically to maximize their impact, and is it possible to measure the impact?

To address employee turnover and to create a framework to answer these questions, a data driven approach is required. Using a data file from kaggle (IBM Data Scientists 2021) that simulates employee data that would be used to analyze attrition, several modeling techniques were developed to predict which employees will leave. The best performing models provide significant insight into the employer's unique causes of attrition and allow for a targeted approach to address those causes.

Data Analysis and Visualization

The original dataset contains a total of 1470 records. Of the 1470 employees, 237 (about 16.1%) left the company and 1233 remained. The following fields were included:

Table 1: Data Field Summary

Field	Data Type	Smallest Value	Largest Value
Age	integer	18	60
Attrition	character	No	Yes
BusinessTravel	character	Non-Travel	Travel_Rarely
DailyRate	integer	102	1499
Department	character	Human Resources	Sales
DistanceFromHome	integer	1	29
Education	integer	1	5
EducationField	character	Human Resources	Technical Degree
EmployeeCount	integer	1	1
EmployeeNumber	integer	1	2068
EnvironmentSatisfaction	integer	1	4

Field	Data Type	Smallest Value	Largest Value
Gender	character	Female	Male
HourlyRate	integer	30	100
JobInvolvement	integer	1	4
JobLevel	integer	1	5
JobRole	character	Healthcare Representative	Sales Representative
JobSatisfaction	integer	1	4
MaritalStatus	character	Divorced	Single
MonthlyIncome	integer	1009	19999
MonthlyRate	integer	2094	26999
NumCompaniesWorked	integer	0	9
Over18	character	Y	Y
OverTime	character	No	Yes
PercentSalaryHike	integer	11	25
PerformanceRating	integer	3	4
RelationshipSatisfaction	integer	1	4
StandardHours	integer	80	80
StockOptionLevel	integer	0	3
TotalWorkingYears	integer	0	40
TrainingTimesLastYear	integer	0	6
WorkLifeBalance	integer	1	4
YearsAtCompany	integer	0	40
YearsInCurrentRole	integer	0	18
YearsSinceLastPromotion	integer	0	15
YearsWithCurrManager	integer	0	17

To prepare the file for analysis and modeling, it was checked for missing (NA) and duplicated records and none were found. Multiple fields that contain the same value for all records and duplicative compensation related fields were removed. The HourlyRate and StockOptionLevel compensation fields were retained. The character fields then were updated to factors. After transforming the data, the dataset was split 80/20% into a training dataset with 1175 records and a holdout test dataset with 295 records. An 80/20 split was used because the file contains a relatively small number of records. The split ensures that the test set is large enough that a small number of outlier records would not have a significant impact on the result, and that the training set contains a sufficient amount of data to train the models. After the data transformation, the following fields were included in the analysis and utilized in the prediction models:

Table 2: Data Field Summary

Field	Data Type	Smallest Value	Largest Value
EmployeeNumber	integer	1	2068
Age	integer	18	60
BusinessTravel	factor	Non-Travel	Travel_Rarely
Department	factor	Human Resources	Sales
DistanceFromHome	integer	1	29
Education	integer	1	5
EducationField	factor	Human Resources	Technical Degree
Gender	factor	Female	Male
HourlyRate	integer	30	100
JobInvolvement	integer	1	4
JobLevel	integer	1	5
JobRole	factor	Healthcare Representative	Sales Representative
JobSatisfaction	integer	1	4

Field	Data Type	Smallest Value	Largest Value
MaritalStatus	factor	Divorced	Single
NumCompaniesWorked	integer	0	9
OverTime	factor	No	Yes
PercentSalaryHike	integer	11	25
PerformanceRating	integer	3	4
RelationshipSatisfaction	integer	1	4
StockOptionLevel	integer	0	3
TotalWorkingYears	integer	0	40
TrainingTimesLastYear	integer	0	6
WorkLifeBalance	integer	1	4
YearsAtCompany	integer	0	40
YearsInCurrentRole	integer	0	18
YearsSinceLastPromotion	integer	0	15
YearsWithCurrManager	integer	0	17
Attrition	factor	No	Yes

Data analysis and visualization can be used to confirm that there are trends and differences that appear to be correlated with attrition. For example, we can see that different jobs have different rates of attrition.

Table 3: Attrition Rates by Job Title

JobRole	Leave	Total	Pct_Leave
Healthcare Representative	8	102	0.0784314
Human Resources	10	44	0.2272727
Laboratory Technician	46	199	0.2311558
Manager	3	76	0.0394737
Manufacturing Director	9	120	0.0750000
Research Director	2	66	0.0303030
Research Scientist	42	241	0.1742739
Sales Executive	46	265	0.1735849
Sales Representative	23	62	0.3709677

We're also able to see differences based on the amount of business travel required of the employee.

Table 4: Attrition Rates by Job Title

BusinessTravel	Leave	Total	Pct_Leave
Non-Travel	12	120	0.1000000
Travel_Frequently	53	226	0.2345133
Travel_Rarely	124	829	0.1495778

If we would like to explore combinations of factors, we can create charts to evaluate combinations that are associated with higher attrition. A combination of job title and hourly rate of pay shows us that some jobs appear to have higher attrition rates than others, but pay rate does not appear to be a significant factor.

Employee Status By Job Title and Pay (Hourly)



Rather than laboriously evaluating one or two factors at a time in an attempt to identify which factors may be drivers of employee attrition, a machine learning approach is utilized to perform that task. Multiple machine learning models are evaluated to determine which performs best. The factors with the most predictive value can then be identified by the model.

Machine Learning Model Selection

Machine learning is used to determine if one or more of the data points can be used to predict employee attrition. By using machine learning, it will be possible to determine that a correlation between attrition and one or more of the factors is non-random, and it will be possible to identify the data points that have the most predictive value. In real world terms, the analysis will be used to determine if employee attrition is caused by one or more of the employment factors in the data, and it should help determine which factors cause employees to leave.

The first step is to choose modeling approaches. Following an approach demonstrated in Introduction to Data Science (Irizarry 2019), the first model selected is logistic regression as a baseline model. For additional

models, a decision tree based modeling approaches seem to be a good fit for the problem. An employer specifically would want to know if some unique combination of factors causes employees to be more likely to leave. For example, maybe employees with 5 – 10 years of experience working part-time in sales are more likely than average to leave the company. A decision tree based solution should provide that type of information to an employer interested in addressing complex combinations of factors that cause attrition. Random forest, as covered in Introduction to Data Science, is selected as a second model. To identify a third model that may be capable of outperforming Random Forest, guidance from the Tree-Based Methods chapter of An Introduction to Statistical Learning with Applications in R (James et al. 2021) is used. Boosting is presented as a solution to reduce potential overfitting of data by iteratively adding to the decision tree using residuals from the model, as it's being trained. The approach is presented as slower, but potentially more accurate than a Random Forest approach. Therefore, the Gradient Boosting method that is available in the R caret package is selected as the third model.

Once the machine learning models have been chosen, tuning control parameters must be determined. The employee attrition dataset has a significant class imbalance. Specifically, the model will be used to predict the attrition class, which is only about 16% of the employee population. The caret Package (Kuhn 2024) covers sampling techniques that can be used to address the negative impact to the prediction model from the class imbalance. Down-sampling, up-sampling and hybrid approaches are presented. Down-sampling, or ensuring that a sample used for training randomly selects an equal number of employees from each class is not ideal because the dataset is small and the approach would result in the use of less of the data for training. Up-sampling, or randomly selecting extra records from the attrition class, with replacement, in order to match the size of the non-attrition class also is not ideal. This approach could cause the model to be inaccurately trained by the repeats to use the exact details of those records when making a prediction. Therefore smote, one of the hybrid option was chosen. Smote, or synthetic minority over-sampling technique, uses the values from the smaller class (attrition) to create similar additional artificial records. This approach allows for the full utilization of the available training data and addresses the concern of over-training on duplicated examples. In addition to a sampling method, a train control method must be selected. Because the dataset is small, the train control will use five fold cross-validation, repeated three times. Five fold cross-validation is used to ensure that the sample used to validate is sufficiently large that a small number of outliers won't have a large negative impact. Cross validation is repeated for a similar reason, since just one random split of the training data into 5 parts could be an outlier result.

Machine Learning Results

Using R, the train control parameters are set and the three models are trained.

```
cval_control <- trainControl(method = "repeatedcv", number = 5, repeats = 3, classProbs = TRUE,
                             summaryFunction = twoClassSummary, sampling = "smote")
model_01 <- train(Attrition ~ . - EmployeeNumber, data = att_train, method = "glm",
                  family = "binomial", metric = "AUC", trControl = cval_control)
model_02 <- train(Attrition ~ . - EmployeeNumber, data = att_train, method = "rf",
                  metric = "AUC", trControl = cval_control)
model_03 <- train(Attrition ~ . - EmployeeNumber, data = att_train, method = "gbm",
                  metric = "AUC", trControl = cval_control, verbose = FALSE)
```

After training the models, all three are used to test their predictive ability against the hold-out data.

```
model_01_predictions <- predict(model_01, newdata = att_test)
model_02_predictions <- predict(model_02, newdata = att_test)
model_03_predictions <- predict(model_03, newdata = att_test)
```

The results are then compared to identify the best performing model.

Table 5: Model Prediction Results

Model	Accuracy	Sensitivity	Specificity	Precision	F1	Balanced Accuracy	Prevalence
Logistic Regression (glm)	0.834	0.854	0.830	0.494	0.626	0.842	0.163
Random Forest (rf)	0.858	0.396	0.947	0.594	0.475	0.672	0.163
Gradient Boosting (gbm)	0.885	0.458	0.968	0.733	0.564	0.713	0.163

A confusion matrix for each model also is evaluated. In the confusion matrix tables, the model predictions are the rows and the actual results are the columns.

Table 6: Logistic Regression Model Confusion Matrix

	Yes	No
Yes	41	42
No	7	205

Table 7: Random Forest Model Confusion Matrix

	Yes	No
Yes	19	13
No	29	234

Table 8: Gradient Boosting Model Confusion Matrix

	Yes	No
Yes	22	8
No	26	239

Because the models are predicting the minority class for unbalanced data, an evaluation of the models should focus on sensitivity and specificity more than accuracy. Since most employees (about 84%) do not leave, overall accuracy scores can be misleading. It can be easy to predict that most employees do not leave and attain a high accuracy score. Sensitivity, or the percentage of employees who left the company that the model predicted would leave and specificity, or the percentage of employees who remained with the company that the model predicted would remain, are better indicators of the usefulness of each model. From the confusion matrix tables, we can see that the Logistic Regression model accurately predicted the largest number employees who left, but also over-predicted leaving. The Random Forest and Gradient Boosting models are more balanced, but missed many employees who left. The Gradient Boosting model was the better performing of the two tree based models.

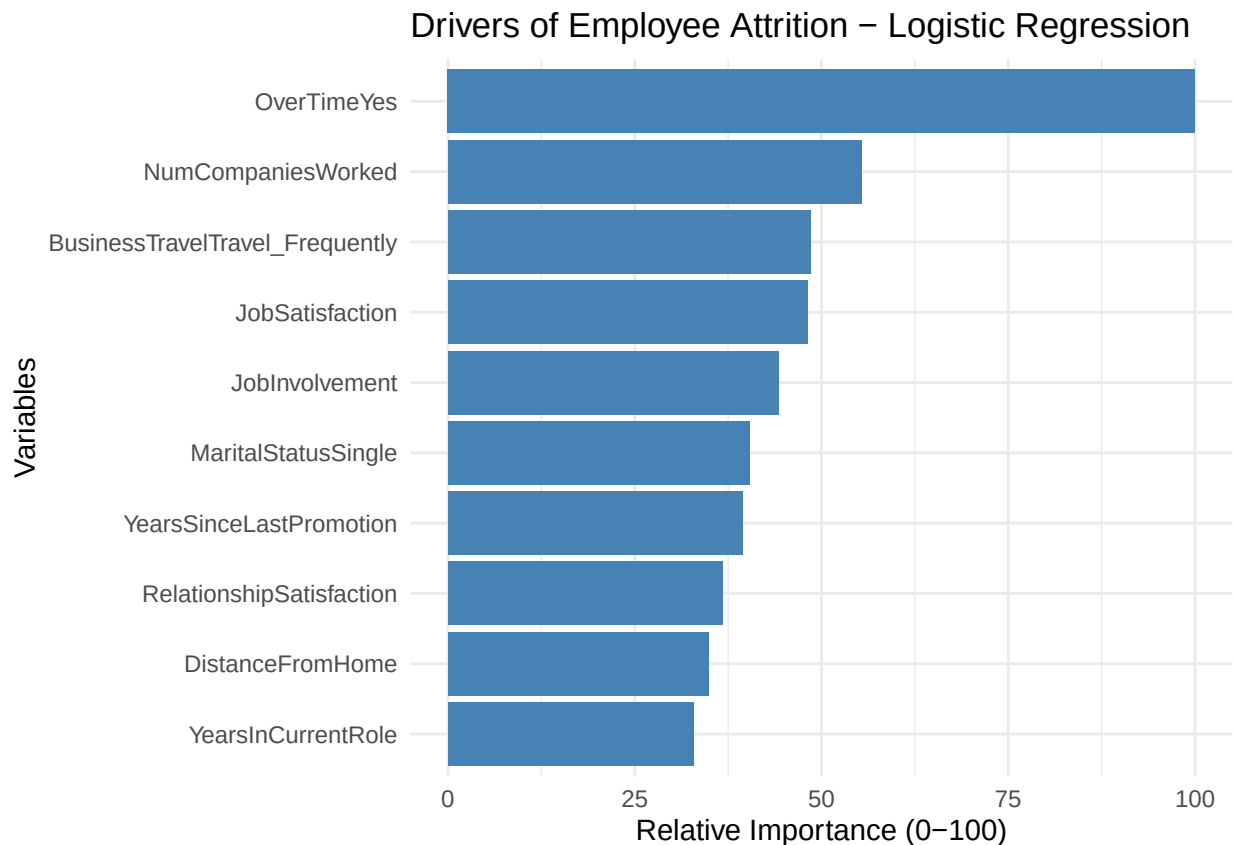
Determination of the best performing model depends on the needs of the use case. In some cases, it may be critical to identify as many positive results as possible, so a larger number of false positives are accepted. In other cases, it may be more useful to have a high level of certainty about a positive prediction, so missing some positive predictions is accepted. In this case, we can use two of the models in a hybrid approach instead of picking just one model. An employee predicted to leave by the Gradient Boosting model was likely to leave (22 out of 30 actually left). The Logistic Regression model accurately predicted a significant proportion of the employees who actually left (41 out of 48), but also predicted that 42 employees who did not leave, would leave. We can combine the results of the two models and categorize each employee as “high risk” if predicted to leave by the Gradient Boosting model, “at risk” if predicted to stay by Gradient Boosting but predicted to leave by Logistic Regression, or “normal risk” if not predicted to leave by either.

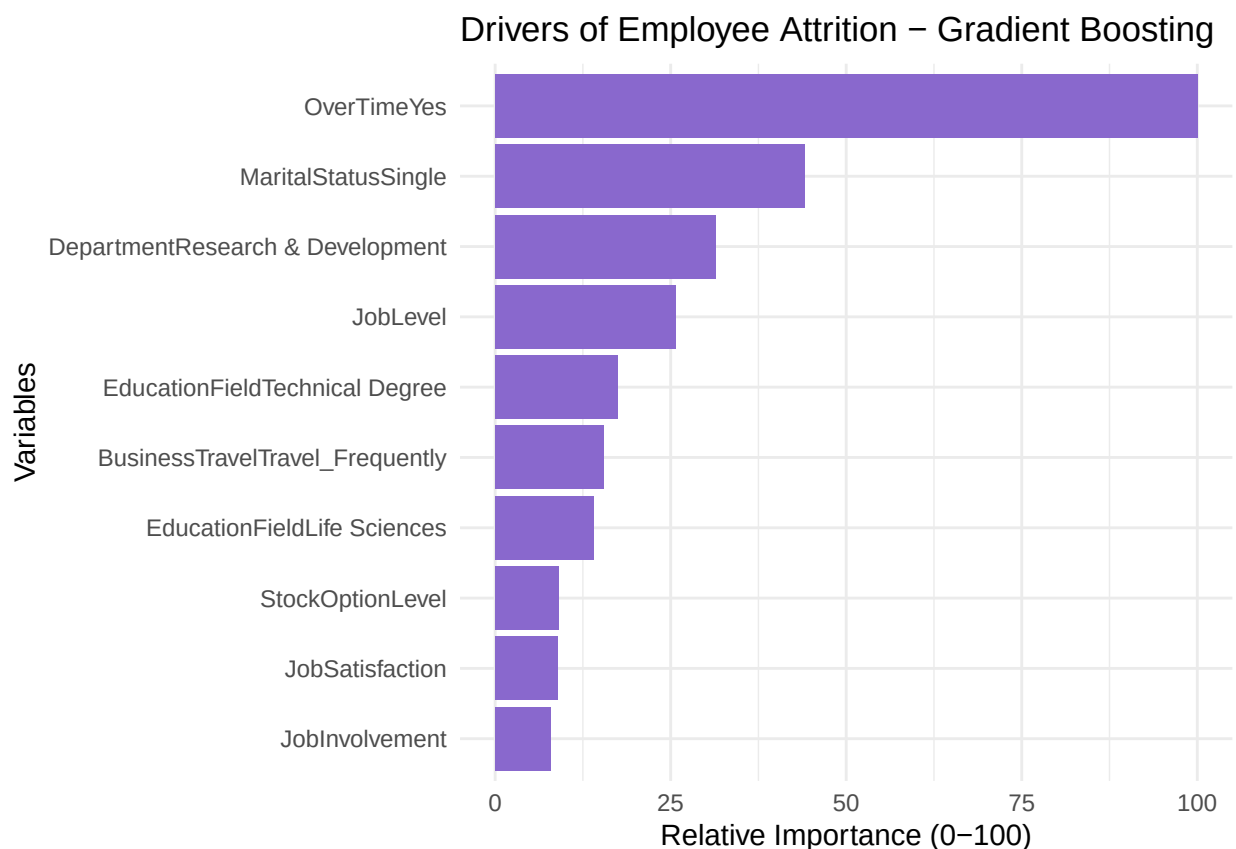
By combining the ability of the Logistic Regression model to identify most employees who leave and the ability of the Gradient Boosting model to be highly accurate when predicting an employee will leave, the company would have been able to categorize employees' attrition risk into three distinct categories. Utilized in this manner, the employees in the hold out test results would have been sorted into the following groups. This sorting would be quite useful to a company wishing to address employment issues with the specific employees who are most likely to leave.

Table 9: Multi-Model Categorization Result

Emp_Category	Leave	Total	Pct_Leave
At Risk	19	57	0.3333333
High Risk	22	30	0.7333333
Normal Risk	7	208	0.0336538

The more important benefit of the prediction models may be to identify the factors used by the models to make their predictions. An employer may wish to evaluate and address the causes of attrition, in an attempt to retain employees. Correlation may not always be cause, but it should be a useful source of information for the employer to consider. In fact, there could be cases where the data provides an insight that employees are unwilling to provide in exit interviews. The top 10 variables used by the Logistic Regression and Gradient Boosting models therefore are extracted for analysis.

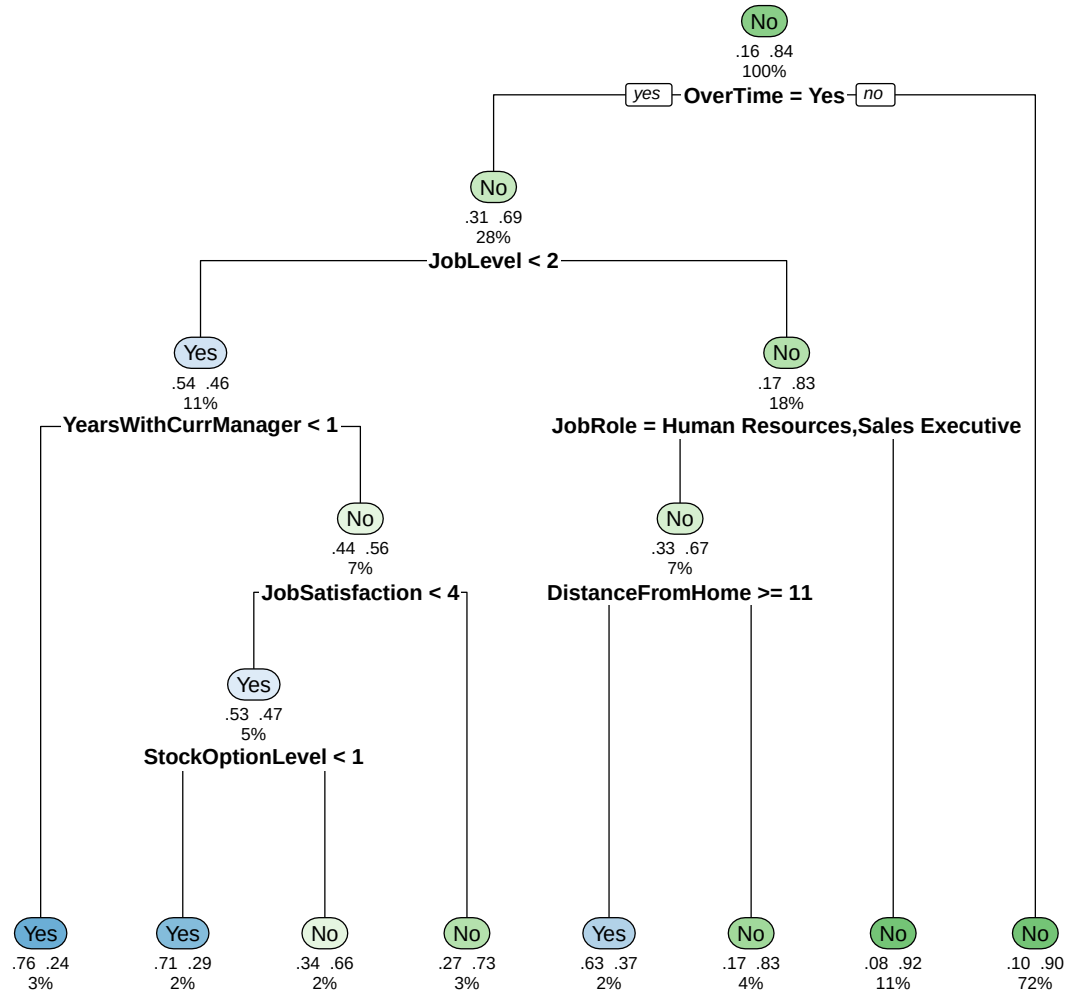




We can see that both models used overtime as the most important predictor, so clearly the employer would want to further explore the number of hours their employees are required to work. The models used some factors that are not actionable, like marital status and the number of companies where the employee has worked, and also used some that could provide additional opportunities, like frequent business travel or working in the Research & Development department.

One additional tool that can be produced for analysis is a decision tree. The decision tree is graphical categorization based on the data. It can provide some insights similar to the insights taken from the prediction models, often for combinations of factors. In this case, the decision tree mirrors the overtime insight already provided by the prediction models, showing that a large proportion of employees (72%) can be grouped into a category where 90% did not work overtime and did not leave. It also provides some additional insight into the employees who did work overtime. For example, if we trace one group of employees through the decision tree, we can see that 11% of the company can be grouped into a category that worked overtime, but were in a job level of 2 or higher, and did not have a Human Resources or Sales Executive role. That group had only an 8% attrition rate, despite the overtime. The combinations of categories in the decision tree could help an employer target more complex combinations of factors that cause employees to leave.

Employee Attrition Risk



Conclusion

The analysis has demonstrated that machine learning can make a significant contribution to an employer's efforts to identify the cause(s) of employee attrition and attempt to address those reasons. Certain steps should be taken by an employer utilizing this approach.

The purpose of this particular machine learning prediction model actually would be to avoid the predicted attrition. If any employer were to take action based on the results of the model, it would impact the model's usefulness in the future. In this example, the model uses overtime as the most important predictor of attrition. If the company were to significantly reduce overtime or otherwise make changes to their overtime policy, they would not want to re-use the model in the future without re-training it, and they would not want to use employment data from prior to the change. A company could follow an iterative process where they periodically collect data and create new predictive models to determine if attrition is correlated with

any employment factors, but they would want to be careful to avoid using data from a past time period when employment conditions were different. The company could repeat the process and address whatever new factor(s) have the highest predictive value, unless or until the model no longer produces statistically significant predictions. Reaching a point where the model no longer is successful at predicting attrition would be a good indication that the company has been able to do as much as they can to address any specific employment conditions that cause employees to leave.

There could be flaws or omissions in the data that would impact the conclusion of causation. Many of the data points, like compensation rates or job roles, are known to the company, but others, like work/life balance or the number of companies where the employee has worked cannot be verified and could be subject to data quality issues. There could also be a confounding factor that is not included in the data, is correlated with the predictors used by the models, and is a more direct cause of attrition. For example, the models rely heavily on overtime to predict attrition. If some other factor like inefficiencies causing work to take longer than necessary or low employee engagement leading to low effort are leading to the overtime that the models used as a predictor, reducing overtime without addressing those issues probably would not achieve the expected improvement in employee retention.

This analysis was performed with a synthetic data file that likely was created intentionally with the correlations that were identified in the analysis. An interesting next step would be to find a way to perform a similar analysis with a larger amount of real world data from a large employer, and to more fully explore many more machine learning models. That work could confirm if the approach and if machine learning technology can be used to predict and mitigate employee attrition in a similar way.

References

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